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Micro-aerated fermentation enhances acetate production from high-rate activated sludge to supply carbon source for heterotrophic denitratation

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ABSTRACT

Heterotrophic denitratation coupled with anammox is a high-efficiency and low-cost process to scavenge residual nitrate and ammonium from the mainstream partial nitritation/anammox process. Fermentate derived from the carbon-concentrated sludge, e.g., chemically enhanced high-rate activated sludge (CEHRAS), is a valuable carbon source for heterotrophic denitratation. However, acetate rather than all volatile fatty acids (VFAs) in the fermentate results in high nitrite accumulation. Micro-aeration may be a promising technology to facilitate VFA production and acetate proportion. This study implemented performance comparison and mechanism analyses on two different micro-aeration modes (i.e., continuous and intermittent micro-aeration) of CEHRAS anaerobic fermentation. Batch fermentation tests showed that the low-intensity micro-aeration (4.62 mL/(L·min)) increased 2.2 times of VFA productions compared with unaerated blank. Higher oxygen dosages (7.69 and 15.4 mL/(L·min)) increased acetate proportion though reduced VFA production, largely due to the elimination of hydrogen inhibition in acetogenesis and the facultative consumption strengthen. Further, the intermittent micro-aeration effectively limits the aerobic respiration in the later stage. Moreover, model substrate tests elucidated that micro-aeration accelerated all steps of protein degradation. The long-term beneficial effects of micro-aeration on VFA production (104 vs. 54.3 mg COD/g VSS) and acetate ratio (59.0% vs. 41.0%) were validated by operating sequencing batch reactors with and without micro-aeration, respectively. Additionally, micro-aeration enriched essential functional microorganisms rather than regulated microbial diversity. Model simulations revealed the crucial roles of increasing acetate proportion in fermentate. The intercellular storage process had higher rate constant and substrate affinity based on acetate than other VFAs, which was the main reason for the higher nitrite accumulation of acetate. Thereby, micro-aeration fermentate from CEHRAS is an ideal carbon source in heterotrophic denitratation.

1. Introduction

In the recent decade, high-rate activated sludge (HRAS) process for carbon capture and anammox-based process for low-cost nitrogen removal are gradually regarded as crucial processes of future water resource recovery factories (WRRFs) [1–3]. In a regular HRAS reactor, the carbon capture efficiency is usually limited by collision efficiency, floc formation, and floc strength. The addition of iron salts could improve these limitation factors, increasing carbon capture efficiency. Moreover, the removal of phosphorus could be enhanced by chemical precipitation [4]. The formed iron-phosphate compounds can be further recovered as vivianite, which gradually becomes a hot recovered type of phosphorus source in recent years [5–7]. The chemically (iron salt) enhanced high-rate activated sludge (CEHRAS) process has the potential to become a more popular carbon capture process than the regular HRAS process in future. Compared with the traditional nitrification/denitrification process, the partial nitritation/anammox (PNA) process has many advantages, such as saving ~60% oxygen and 100% organic demands and reducing ~90% sludge production [8,9]. However, the theoretical maximum nitrogen removal of PNA is limited to 89% due to the oxidization of nitrite to nitrate by the nitrite:nitrate oxidoreductase on anammoxosome [10]. Meanwhile, the possible low ratio of alkalinity to

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ammonium could limit the conversion of ammonium to nitrite in the PN process, resulting in a lower nitrite to ammonium ratio than that of theory and high concentration of residual ammonium in the effluent [11]. Moreover, the complete suppression of nitrite-oxidizing bacteria (NOB) is difficult in practice [12]. In these cases, the effluent total nitrogen (TN) concentration of PNA is hard to meet the gradually strict discharge standards, especially treating high ammonium concentration of wastewater, such as toilet wastewater (blackwater) and anaerobically digested sludge liquor.

Heterotrophic denitratation, the reduction of nitrate to nitrite with organics as electron donor, is an effective remedy to the problem of high-concentration inorganic nitrogen in PNA effluent. By heterotrophic denitratation, the nitrate formed in the anammox reaction could be reduced to nitrite, which could be reused by anammox bacteria to convert the residual ammonium to dinitrogen [13–16]. Thus, most of the residual nitrate and ammonium could be removed. However, after carbon capture by CEHRAS process and aerobic oxidation by the partial nitritation process, there is little organic source in mainstream for heterotrophic denitratation [4]. Anaerobic fermentation is a promising technology to recover the carbon source captured and concentrated in CEHRAS. The produced volatile fatty acids (VFAs) can serve as the organic source for heterotrophic denitratation. However, not all VFA components perform well in terms of nitrite accumulation [17–19]. As the carbon source, acetate can result in a higher nitrite accumulation ratio than other VFA components and is usually adopted in lab-scale and pilot-scale reactors of heterotrophic denitratation [12,20–22]. Therefore, it is of great significance to increase acetate production and proportion of the fermentation of CEHRAS.

Some advanced oxidizing agents, such as potassium permanganate [23], peroxymonosulfate (PMS) [24], potassium ferrate [25,26], hydrogen peroxide [27], calcium hypochlorite [28], and calcium peroxide [29,30], were found conducive to the transformations of other VFAs to acetate. Zhang et al. [28] found that the reactive oxygen species (mainly ·OH) induced by $Ca(ClO)_2$ (0 vs. 0.05 g/g TSS) might contribute to the improvement of acetate proportion (30.6% vs. 46.3%). Li et al. [29] demonstrated that the ratio of acetic acid was enhanced from 45.1% to 60.2% with CaO_2 addition increasing from 0 to 0.2 g/g VSS, while $Ca(OH)_2$ had no obvious effect on the ratio of acetate, and assumed that this phenomenon might be attributed to the chemical oxidation of CaO_2 ($\cdot O_2^-$ and ·OH) or functional microbe change, i.e., the increasing population of Firmicutes. Yang et al. [24] and Xu et al. [23] reported that the PMS and $KMnO_4$ addition facilitated the acetate production and the removal of larger-molecule VFAs, respectively. The behind mechanisms were both that all the bioprocesses and enzymes relevant to anaerobic fermentation were inhibited in the presence of powerful oxidants (PMS and $KMnO_4$), but the suppression to methanogenesis and F420 were much severer than that to others, such as acetogenesis process and acetate kinase. It is probable that the inhibition effects on all bioprocesses increase the fermentation period or reactor size. Moreover, the addition of chemicals has other obvious disadvantages, such as the high dosage increasing operating costs and the chemical wastes production, and the oxidative property increasing the risks of transportation, preservation, and use.

Compared to the above-mentioned chemicals, oxygen is a more environmental-friendly oxidant, which widely exists in the atmosphere with a high proportion. Recently, micro-aeration, supplying a small amount of pure oxygen gas or air, has been considered a cheaper green substitute for the dosing of advanced oxidizing agents. Although the aim of micro-aeration in most literature is to decrease VFA accumulation and increase CH_4 production of anaerobic digestion [31–34], the mechanisms behind micro-aeration assisting anaerobic digestion can be used to increase the acetate production and proportion of anaerobic fermentation. The underlying rationales are as follows: (1) oxygen is an electron acceptor of facultative microbes and can augment their activity and diversity [35,36]; (2) the reactive oxygen species stimulated by oxygen can degrade the phospholipid membrane, DNA, and protein [37,38]; (3) the hydrogenotrophic methanogens can converse the CO_2 produced from the aerobic oxidation of VFA to methane [32]; (4) the co-existence interaction between microorganisms can be established, for instance, producing anti-oxidative enzymes and protecting inner oxygen-sensitive anaerobes (e.g., methanogens), however, over-dose oxygen beyond the oxygen consumption capacity of facultative microorganisms would break this aerotolerant mechanism [39]. The first two mechanisms were conducive to the hydrolysis and acidogenesis processes of anaerobic fermentation. The third mechanism indicates that oxygen can enhance the aerobic oxidation of VFAs by the aerobic respiration of facultative microorganisms. The fourth mechanism implies that the conversion of acetate to CH_4 could be suppressed under higher-dose oxygen conditions. Thus, micro-aeration is a switching tool among the multiple purposes for the anaerobic processes. Yu et al. [40] proposed that the low-dose aeration (5 mg O_2/L/Cycle) indeed enhanced the hydrolysis efficiency (34.7% vs. 48.7%) and methane yield of blackwater (39.6% vs. 50.7%), while the VFA accumulation sharply increased and the methane yield simultaneously declined at higher oxygen dosages (50 and 150 mg O_2/L/Cycle). The medium-dose aeration (50 mg O_2/L/Cycle) obtained maximum VFA accumulation. Sawatdeenarunat et al. [41] reported that the positive effects of micro-aeration on anaerobic digestion of lignocellulosic biomass, and also found VFAs yield showed a quadratic correlation with oxygen dosage.

Based on the above literature review and further analyses, micro-aeration is considered a promising and feasible method for improving CEHRAS anaerobic fermentation, especially in enhancing acetate production. However, several vital problems are worth to be clarified: (1) which is the better micro-aeration mode, continuous or intermittent micro-aeration? (2) the effects of micro-aeration on each process in anaerobic fermentation are unclear until now; (3) the performance of micro-aerated fermentate derived from CEHRAS as a carbon source for the heterotrophic denitratation has not been verified.

To our best knowledge, this study is the first report to use micro-aeration to increase the acetate production and proportion of anaerobic fermentation with CEHRAS as substrates. This study compared the performance of the continuous and intermittent micro-aeration strategies. Batch fermentation experiments investigated the influences of aeration intensity and aeration cycle time and then explored the enhancement mechanism of acetate production by model substrate experiments. Further, sequencing batch reactors with or without micro-aeration fed with actual CEHRAS were carried out to validate the long-term effects of micro-aeration on VFA production and distribution. Then, batch denitrification experiments verified the high nitrite accumulation performance of micro-aerated fermentate. Finally, the influences of carbon/nitrogen ratio, acetate proportion in fermentate, and process parameters on heterotrophic denitratation were predicted by modeling.

2. Method and material

2.1. Source of CEHRAS

The sludge for fermentation was collected from a lab-scale CEHRAS sequencing batch reactor (SBR) with a working volume of 4 L. The CEHRAS SBR was fed on real blackwater collected from a maintenance hole connecting toilets of one campus library and two teaching buildings on the Siping Campus of Tongji University, Shanghai City, China. Characteristics of the real blackwater were summarized in our pervious study [11]. The total sludge retention time (SRT) and aerobic hydraulic retention time (HRT) of CEHRAS SBR were 0.25 ~ 0.5 days and 30 min, respectively. In order to enhance the flocculation process, a total of 60 mg $FeCl_3$ (calculated as Fe) was added with 2 L influent of real blackwater. The CEHRAS was thickened and stored in a refrigerator at 4 ℃. The solid concentration of the thickened CEHRAS was 10 ~ 15 g VSS/L. Before use, the CEHRAS in different storage buckets was mixed into a container and diluted to 9.0 g VSS/L.

2.2. Lab-scale fermentation reactor

The lab-scale fermentation reactor was a double-layer glass vessel with a working volume of 1.0 ~ 1.5 L (Fig. 1). The out layer was a water bath that could exchange heat with a self-made thermostat (a module of four semiconductor chilling plates (CL-C067, Heke Tech, China)), which could be switched between heating and cooling states by exchanging their positive and negative poles connecting power supply. The reaction liquid in the inner layer was mixed by a 50 W overhead stirrer (LC-OES-60SH, LICHEN, China). Compressed gas (nitrogen gas or air) was injected into the reaction liquid bottom by a stainless ring sparger with evenly distributed holes. The gas flowrates were manually regulated by rotor flowmeters. The fermentation reactor was air-tight. However, the injected gas could leave the reactor by a one-way valve. The fermentation reactor equipped with a pH sensor (SIN-PH-5015, LONTROL, China) and an oxidation–reduction potential (ORP) sensor (LONTROL, China). The pH sensor also detected temperature (T). The pH, ORP and T signals were logged by the self-made control system. The control system equipped with an S7-200 smart programmable logical controller (PLC) (SIEMENS, Germany). The pH and T of reaction liquid and aeration frequency were automatically controlled by the control system. According to the buffer capacity of reaction liquid, different concentration of HCl (0.2 ~ 2 mol/L) and NaOH (0.2 ~ 2 mol/L) solutions were prepared for pH control.

All the batch fermentation and denitrification experiments and the long-term fermentation experiments could be executed in the lab-scale fermentation reactors. Four lab-scale fermentation reactors were used in this study.

2.3. Batch fermentation experiments of continuous micro-aeration

Firstly, 1.3 L CEHRAS was filled into a lab-scale fermentation reactor, and the CEHRAS was stirred at 200 rpm. Then, compressed nitrogen gas (1 L/min) was injected for 10 min to strip potential dissolved oxygen (DO). Next, pH and T controls were started. When pH and T values reached their corresponding setpoints (pH = 8.0, T = 25 ℃, respectively), continuous micro-aeration was executed: Compressed air was continuously supplied to the fermentation reactor until the batch experiment was over (one week usually). The pH was set closely to the actual pH of the original CEHRAS, mainly for decreasing the addition amounts of chemical reagents as far as possible. Four micro-aeration intensities were set: 0 (control), 4.62, 7.69, and 15.4 mL air per minute per liter sludge (mL/(L·min)). Every 0.5 ~ 2 days, 20 mL sample was grabbed using a syringe. One 10 mL sample was immediately centrifuged at 4000 rpm for 5 min. Then, the supernatant was filtrated by 0.45 μm polyethersulfone (PES) filters. The filtrate and another 10 mL sample were immediately stored at 4 ℃ waiting for analyses.

2.4. Batch fermentation experiments of intermittent micro-aeration

The basic procedures of batch fermentation experiments with intermittent micro-aeration were the same as those with continuous micro-aeration. The difference was that the compressed air supply of the intermittent micro-aeration was periodically switched on and switched off during the fermentation period. One intermittent aeration cycle was comprised of an aeration phase (air supply on) and a non-aeration phase (air supply off). In this study, the total cycle time was fixed at 60 min, while the aeration cycle time was adjusted for different experimental groups. Three aeration cycle times were implemented, i.e., 15 min, 30 min, and 45 min. The micro-aeration intensity during the aeration phase was controlled at 7.69 mL/(L·min) in experimental groups. The continuous micro-aeration batch with 7.69 mL/(L·min) in section 2.3 was regarded as the control group, which could be considered as intermittent aeration with 60 min of aeration phase cycle time. The "15 min/60 min", "30 min/60 min", "45 min/60 min", and the "60 min/60 min" were used to represent the above experimental and control groups in the following text in brief.

2.5. Long-term operation of intermittent micro-aeration fermentation

One lab-scale fermentation reactor (section 2.2) worked with sequencing batch procedure was implemented to assess the long-term operation performance of CEHRAS fermentation with intermittent micro-aeration assistant. Intermittent aeration was conducted during

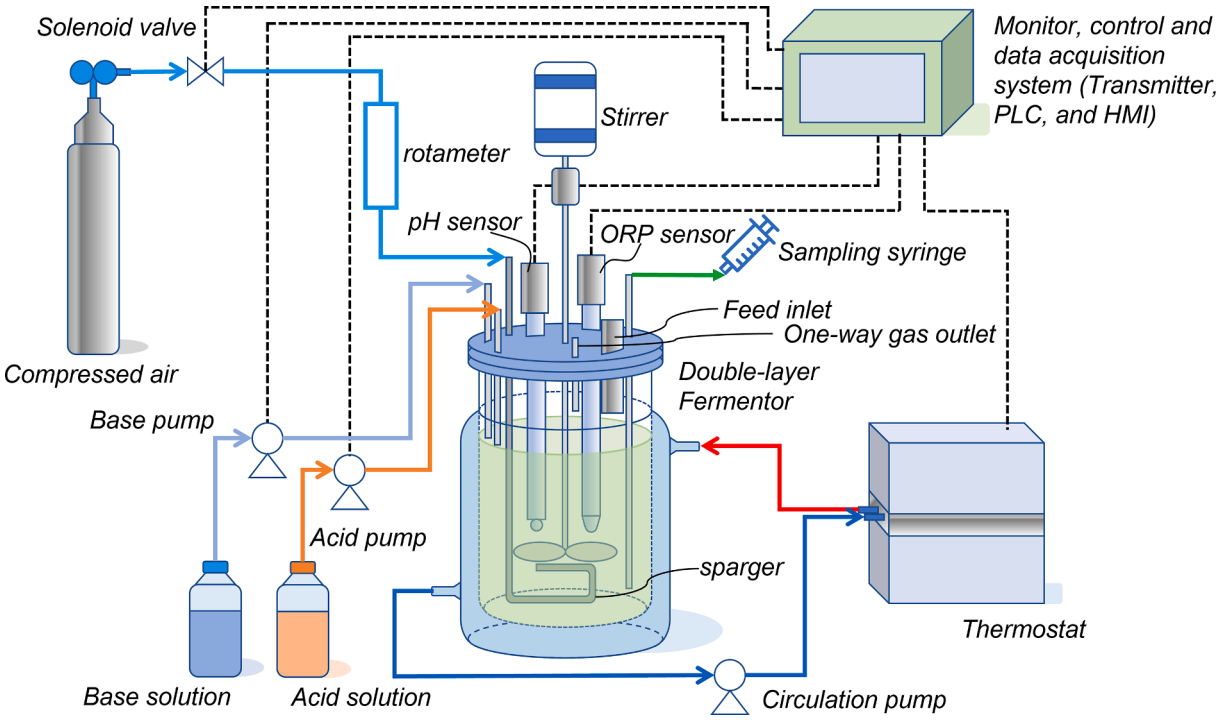


Fig. 1. Schematic diagram of the lab-scale fermentation reactor.

the whole fermentation period. Herein, we used "MaSBR" to represent this reactor. The total intermittent aeration cycle was 60 min with an aeration cycle of 45 min. The mixed liquid volume was 1.3 L, while the HRT and SRT were both 5 days, which means a total of 260 mL fermentate was manually discharged and replaced with fresh CEHRAS every day. In addition, a same lab-scale fermentation reactor without aeration (AnSBR) was also operated in control. The daily replaced fermentate volume was 650 mL, resulting in 2-day HRT and SRT. According to the rule of thumb, i.e., a period of 2 ~ 3 times of SRT is required to reach the quasi-steady state, the operational periods of AnSBR (SRT = 2 days) and MaSBR (SRT = 5 days) were 15 days and 30 days, respectively. The pH and T of both reactors were controlled at 8.0 and 25.0 ℃, respectively.

2.6. Influences of micro-aeration on hydrolysis and acidification in anaerobic fermentation

Anaerobic fermentation of sludge generally contained several basic steps, i.e., hydrolysis, acidogenesis, acetogenesis, and these steps directly affect the eventually VFA production. Thus, it is essential to assess the role of micro-aeration on each step involved in anaerobic fermentation. However, these steps generally co-occur in actual sludge fermentation, for instance, the soluble chemical oxygen demand (sCOD), soluble protein, soluble carbohydrate and three-dimensional excitation emission matrix (3D-EEM) on 2 d only reflect the comprehensive effect of solubilization and hydrolysis. Herein, the batch experiments of influence of micro-aeration intensity (0 ~ 15.4 mL/(L·min)) on above processes were performed with model substrates.

Hydrolysis tests: 1 g/L dextran (model carbohydrate), 2 g/L bovine serum albumin (BSA, model protein) and inoculum were added in the synthetic wastewater. The synthetic wastewater compositions were according to Wang et al. [42]. Because CEHARS also contains high concentration of substrate with low biological activity, the inoculum was a little of wasted activated sludge (WAS), which was aerated overnight to remove unused substrate. In addition, the aeration system was start up when ORP of the fermentation systems was lower than −200 ~ −150 mV for preventing the direct oxidation of model substrates in initial periods. The hydrolysis degree was evaluated by the degradation rate of dextran and BSA after 3 days.

Acidification tests: all operations were the same as Hydrolysis Test except that the model substrates were replaced with 1 g/L glucose (model monosaccharide) and 2 g/L L-glutamate (model amino acid). The acidification degree was evaluated by the degradation rate of glucose and L-glutamate after 3 days.

2.7. Denitrification batch experiments

All denitrification batch experiments were performed in a lab-scale fermentation reactor with a working volume of 1.5 L (section 2.2). Additional DO sensor (COS22D, Endress + Hauser, Germany) was used to monitor DO in the bulk. The sludge for denitrification tests was taken from the membrane tank of a municipal wastewater treatment plant (WWTP) located in Hangzhou City, China. The sludge was washed three times to decrease residual substrate concentration, and the sludge concentration was finally all diluted to 2.5 g VSS/L in the reactor. Meanwhile, 2.8 mL macro-element stock solution and 0.42 mL trace-element stock solution was added into the reactor [43]. The T and pH were both controlled at the optimal denitratation conditions (pH = 8.5 ± 0.1, T = 25.0 ± 0.1 ℃, see in Fig. S1 and Fig. S2), and the reactor was stirred at 200 rpm. The sludge mixed liquid was purged with nitrogen gas until the DO was lower than 0.05 mg/L.

While pH, T, and DO reached and ensured stable initial conditions, NO_3-N stock solution (25 mg N/mL) was added into the reactor to achieve the initial NO_3-N concentration of 30 mg N/L. Then, different carbon source was added, resulting in an initial COD/NO_3-N ratio of 3.0 or 6.0 mg COD/mg N. The carbon sources were sodium acetate solution (170 mg COD/mL), synthetic fermentates with the different acetate proportion (0%, 20%, 40%, 60%, 70%, 80%, and 100%) and real micro-aeration fermentate from the MaSBR. During the 240 min batch period, 10 mL sample was grabbed with a syringe every 20 ~ 30 min. The sample was injected into a 15 mL centrifuge tube, and 20 μL chloroform was added to completely inhibit biological activity. Concentrated HCl and H_2SO_4 were not recommended, because they caused significant interference on chemical analysis. The sample was immediately centrifuged at 4000 rpm and then filtered with 0.45 μm PES filters. The filtrate was stored at 4 ℃ before use.

2.8. The establishment of dual-substrate and two-step denitrification model

The dual-substrate and two-step denitrification (DSTSD) model is a simplified modification of the Activated Sludge Model 3 (ASM3). The DSTSD model consists of five components and six transformation processes. The dual substrates are acetate (S_{Ac}) and non-acetate VFAs (S_{NAc}). The two-step denitrification means that nitrate (S_{NO3}) and nitrite (S_{NO2}) are separately considered. In the DSTSD model, only anoxic storage and growth processes of denitrifiers were considered. The processes and their kinetic rate expressions are listed in Table S1. The stoichiometric and composition matrix is shown in Table S2.

2.9. Analytical methods and statistical analysis

The procedures of DNA extraction and 16S rRNA gene high-throughput sequencing and analytic methods of chemical and physical composition are detailed in the Supporting Information. The difference between a group of data and a reference value was compared by one-sample t-test. The difference between two groups of data was compared by two-sample t-test. For two-sample t-test, if the variance was not equal, Welch correction was used. Correlation coefficients were calculated by Pearson method. All the statistical analyses were conducted on the platform of OriginPro 2018.

3. Results and discussion

3.1. Influences of aeration intensity on CEHRAS anaerobic fermentation with continuous micro-aeration

3.1.1. VFAs production and compositions

Figure 2(a) presents the productions of VFAs during seven-day anaerobic fermentation with and without continuous micro-aeration. Obviously, the VFA production of high-rate activated sludge was enhanced by continuous micro-aeration, of which efficacy was dependent on the aeration intensity. The VFA production of the unaerated batch reached the maximum at 40 h, but those of the aerated batches still showed a considerable increase thereafter. The maximum VFA productions were 113, 97.6, and 73.8 mg COD/g VSS, respectively, which were 2.2, 1.9, and 1.5 times higher than that of the unaerated blank (i.e., 50.6 mg COD/g VSS), respectively, when the aeration intensities were 4.62 mL/(L·min), 7.69 mL/(L·min) and 15.4 mL/(L·min), respectively. Wang et al. [44] also reported that the maximum VFAs production enhanced by 54.3% after micro-oxygen bio-pretreatment with oilseed rape straw as substrate. It suggests that a trace injection of oxygen probably promote the propagation of facultative anaerobes and the synthesis of extracellular hydrolases, consequently accelerating hydrolysis and acidogenesis processes. The decreasing trend of the maximum VFA production with increased micro-aeration intensity might be due to the enhancement of aerobic respiration by facultative microorganisms [39]. In addition, the VFAs did not decrease in the later stage of the unaerated batch, indicating that the absence of methanogenesis process due to the short SRT (0.5 days) of high-rate activated sludge, which is accordant with scarcely methane in gas composition detection of all groups. The non-existence of methanogens (see in

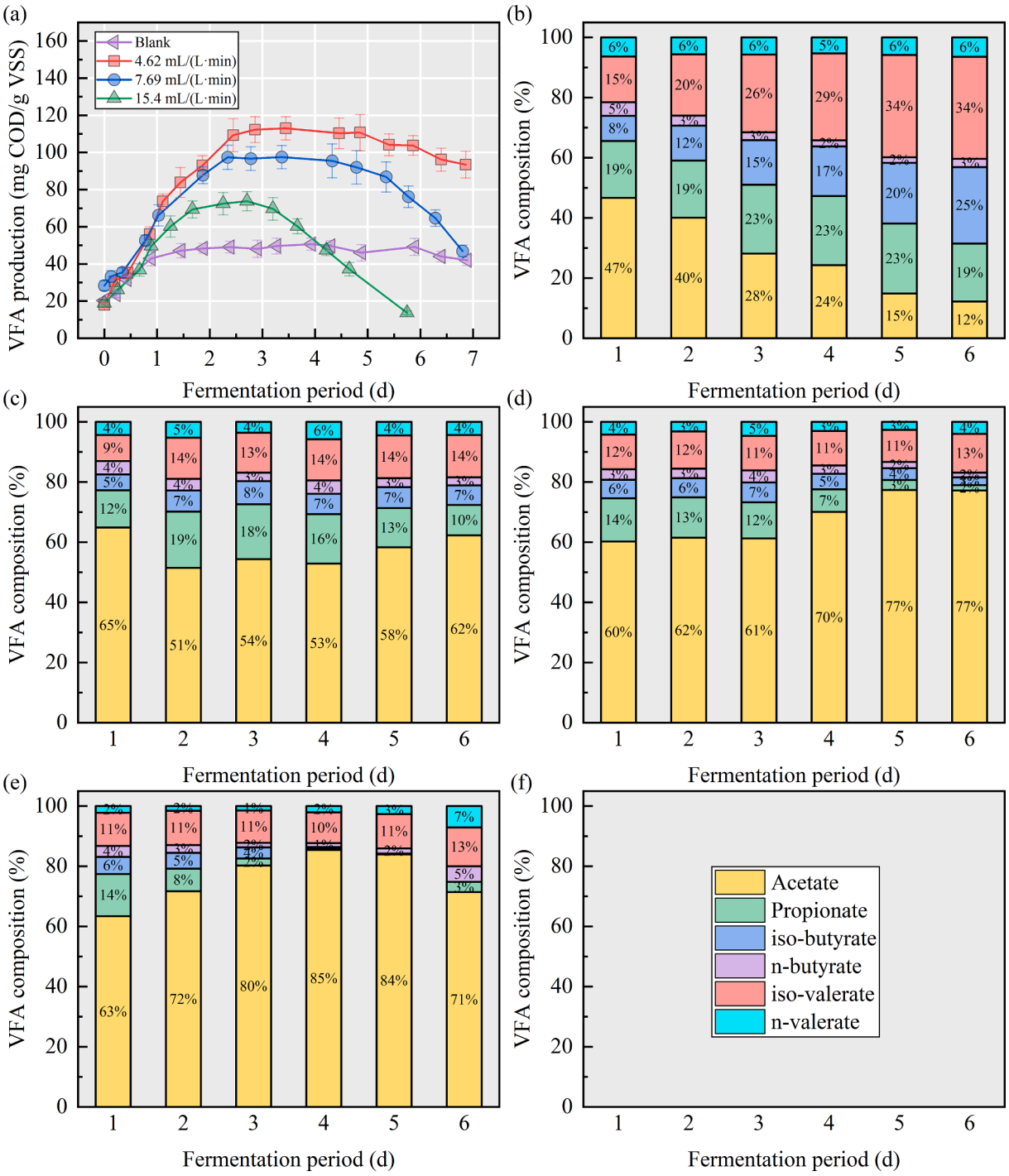


Fig. 2. Total VFAs production profiles with continuous micro-aeration during whole CEHRAS anaerobic fermentation (a) and VFAs distribution at different aeration intensity levels: blank (b), 4.62 mL/(L·min) (c), 7.69 mL/(L·min) (d), and 15.4 mL/(L·min) (e). Error bars in (a) represent standard errors of triplicate samples.

section 3.3.3) is a unique advantage of CEHRAS, comparing to the secondary clarifier sludge with a long SRT. In the long-SRT wastewater treatment plants, the anaerobic environments in anaerobic tanks, the bottom of clarifier and sludge pipelines could support the growth of methanogens.

Figure 2(b)~(f) show the VFA distributions during the fermentation period. For the unaerated batch, acetate and propionate were dominant in the initial stage (i.e., the first 2 ~ 3 days). As the fermentation period prolonged from day 3 to day 6, *iso*-butyrate and *iso*-valerate were the predominant VFAs, subsequently. Note that propionate accounted for approximately 20% of the total VFAs during the whole fermentation period. Acetate proportion remarkably improved after micro-aeration treatment. As the aeration intensity increased from 4.62 mL/(L·min) to 7.69 mL/(L·min) and 15.4 mL/(L·min), the acetate proportion on day 4 increased from 53% to 70% and 85%, respectively, significantly (one-sample *t*-test, $p = 0.039$) higher than the 24% of the unaerated batch. In contrast, the ratio of propionate declined apparently. As seen in Fig. 2 (e), propionate was consumed completely after three days when the aeration intensity increased to 15.4 mL/(L·min). In addition, *iso*-butyrate and *iso*-valerate also decreased to some extent in the presence of micro-aeration. Therefore, the micro-aeration is beneficial to the transformation of other VFAs to acetate, i.e., the acetogenesis process. In

general, acetogenic bacteria produce hydrogen, which inhibits the acetogenesis process itself instead [45]. In micro-aeration conditions, hydrogen could be utilized as the electron donor by some microorganisms, making the hydrogen pressure low enough for the acetogenesis process [32]. In addition, the hydrogen might be expelled from the fermenter by the air supply, which would play a significant role in reducing the hydrogen pressure.

Notably, the VFA distribution greatly influences the downstream application of VFAs. It is known that acetate and propionate were the suitable carbon source in terms of the traditional denitrification [46]. As for the heterotrophic denitratation process, acetate caused higher nitrite accumulation performance compared to other VFAs [12]. Rijn et al. [19] found that longer-chain VFAs (such as butyrate and valerate) did not present obvious nitrite accumulation, mainly owing to the tendency of donating electrons to nitrite reductase. Ali et al. [16] observed the nitrite accumulation ratio (NAR) of > 80% using pure acetate as a carbon source in long-term pilot operation, whereas only 49% was achieved with primary sludge fermentate (50% acetate proportion). Thereby, high proportion acetate of fermentate from CEHRAS would be a preferred substitute carbon source in the heterotrophic denitratation system.

3.1.2. *Influences of micro-aeration on solubilization, hydrolysis, acidogenesis and acetogenesis processes*

The concentrations of soluble proteins, soluble carbohydrates and sCOD after two-day micro-aeration fermentation were measured to evaluate the solubilization and hydrolysis processes of CEHRAS. As presented in Fig. 3(a), there were no large differences in the soluble carbohydrates among different batches. The concentrations of soluble carbohydrates were respectively 31.9, 43.4, 41.5, and 35.5 mg/L in batches with aeration intensity of 0, 4.62, 7.69, and 15.4 mL/(L·min), respectively. In contrast, the contents of released proteins were obviously higher than that of carbohydrates, and the variation trend of which was consistent with VFA productions among different levels of micro-aeration intensity. The concentrations of soluble proteins were 172, 145, and 122 mg/L, respectively, when the micro-aeration intensities were controlled at 4.62, 7.69, and 15.4 mL/(L·min), respectively, much higher than that of the unaerated batch (i.e., 93.9 mg/L). There are two possible reasons for the similar contents of soluble carbohydrates in Fig. 3(a). First, the hydrolysis and acidification of carbohydrates were rapid and unrelated to the aeration intensity basically (see Fig. 3(g), (h)). In this case, the released carbohydrates and the corresponding hydrolyzed products (i.e., oligosaccharides) could be readily consumed. Second, the contents of carbohydrates in CEHRAS were much lower than proteins. Moreover, the above results imply that the impacts of micro-aeration intensity on hydrolysis and acidification of proteins play a decisive role in VFA production. Apparently, the solubilization performance was dramatically improved with the presence of micro-aeration, which was also reported by Lim and Wang [31], and then the considerable released macromolecular proteins would be degraded in the hydrolysis. Extensive studies have illustrated that the beneficial effects of micro-aeration treatment on hydrolysis process [31,40,47–49], and the hydrolysis rate might be positively correlated with the oxygen injection amount. Therefore, in the range of 4.62 ~ 15.4 mL/(L·min), the soluble protein decreased with the increased aeration intensity (Pearson correlation coefficient = -0.958). Meanwhile, the aerobic respiration by facultative bacteria could be strengthened with the increased oxygen injection amount. In this case, more amino acids would be converted to CO_2 and H_2O rather than VFA. Accordingly, VFA, sCOD, and soluble protein decreased simultaneously at higher aeration intensity in Fig. 2(a), Fig. 3(a) and (b).

The EEM fluorescence spectrum was employed for further characterizing dissolved organic matters (DOMs) in supernatants of the micro-aerated and unaerated batches. Two protein-like substance peaks were observed in the EEM fluorescence spectrum, i.e., peak A (Ex/Em 270/300) and peak B (Ex/Em 275/350), which were associated with tyrosine and tryptophan protein-like substances, respectively. Peak C ((Ex/Em > 280/ > 380) was related to the humic acid-like organics [50–52].

The intensities of peak A and peak B in the fermenter with micro-aeration treatment were greater than those of the unaerated batch. However, peak A and peak B gradually became weak as the aeration intensity increased. It could be inferred that micro-aeration can accelerate the release of DOMs (mainly are tryptophan and tyrosine protein-like substances). Nevertheless, if the micro-aeration intensity is too high, the hydrolysis rate of protein might be sped up, and then remarkable amino acids tend to be utilized by the aerobic respiration of heterotrophic bacteria, which was consistent with the results inferred from proteins variation in Fig. 3(a). Additionally, compared to the unaerated batch, peak C attributed to the humic acid-like organics only appeared in the three micro-aerated batches. More specially, the intensities of peak C in the fermenter with high micro-aeration intensities (7.69 and 15.4 mL/(L·min)) were lower than that of the low micro-aeration intensity (4.62 mL/(L·min)). It indicates that the massive humic acid-like organics released to the supernatant with the accelerate of sludge disintegration triggered by micro-aeration. Previous studies confirmed that recalcitrant organics could be partially degraded to biodegradable substrates after PMS or $KMnO_4$ addition. It was reported that $KMnO_4$ could destroy organics containing C = C bonds or electron-rich moieties, while the PMS could scavenge conduction band electrons and form sulfate radical [23,24]. Furthermore, the amounts of intracellular enzymes (e.g., monooxygenase and peroxidase) are abundantly secreted in the presence of limited aeration, causing the activation and structure change of refractory organics [38,53]. Similarly, part of recalcitrant organics could be degraded and used under micro-aeration condition, which might provide more biodegradable organics for VFA generation. However, the biodegradable organics derived from the humic acid-like organics were mostly used by aerobic respiration rather than anaerobic fermentation with higher micro-aeration intensities (7.69 and 15.4 mL/(L·min)), according to the decreased VFAs in the later stage of the fermentation period.

From the above results, the micro-aeration participation is effective in enhancing CEHRAS disintegration and VFAs production. The anaerobic fermentation of CEHRAS containing several bioprocesses, i.e., hydrolysis, acidogenesis, acetogenesis, more importantly, the VFA-consumption steps (aerobic respiration by facultative anaerobes) co-occur in the micro-aerated fermentation, these multiple steps could not be readily differentiated. Therefore, model substrates were utilized to figure out the role of micro-aeration on each fermentation step. Fig. 3(g) presents the influence of different levels of micro-aeration on the hydrolysis of BSA (model protein) and dextran (model carbohydrate) on day 3. Dextran showed a significantly (two-sample *t*-test, Welch correction, $p = 0.003$) higher degradation than BSA, suggesting that microbes preferred carbohydrate to protein. However, the degradation rates of dextran of all batches were >98%, and there was no significant (one-sample *t*-test, $p = 0.725$) difference between the blank batch and the micro-aerated batch, consistent with the soluble carbohydrate results in Fig. 3(a). In contrast, the degradation efficiency of BSA was 11.4%, 31.8%, 30.4%, and 46.5%, respectively, when the micro-aeration intensities were 0, 4.62, 7.69, and 15.4 mL/(L·min). Thus, the micro-aeration treatment could facilitate BSA hydrolysis, implying that the micro-aeration treatment stimulates the synthesis and/or release of proteases by fermentative bacteria. Ruan et al. [33] also reported that the amounts of protease could increase from 1.8 to 4.01 U/g after micro-aeration pretreatment in sludge anaerobic digestion (AD) process. The other potential promotion mechanisms of hydrolysis rate in literatures are as follows: (1) Facultative bacteria produce higher energy from the organic degradation in micro-aerobic environment, resulting in higher specific growth rate and biomass yield, thereby promoting degradation of the slowly biodegradable compounds [31,39]. (2) Oxygen molecules partially are reduced to reactive oxygen species (i.e., O_2^-, H_2O_2, and ·OH) and cause cell membrane destruction, protein and DNA breakage [38]. (3) The diversity and/or activity of facultative

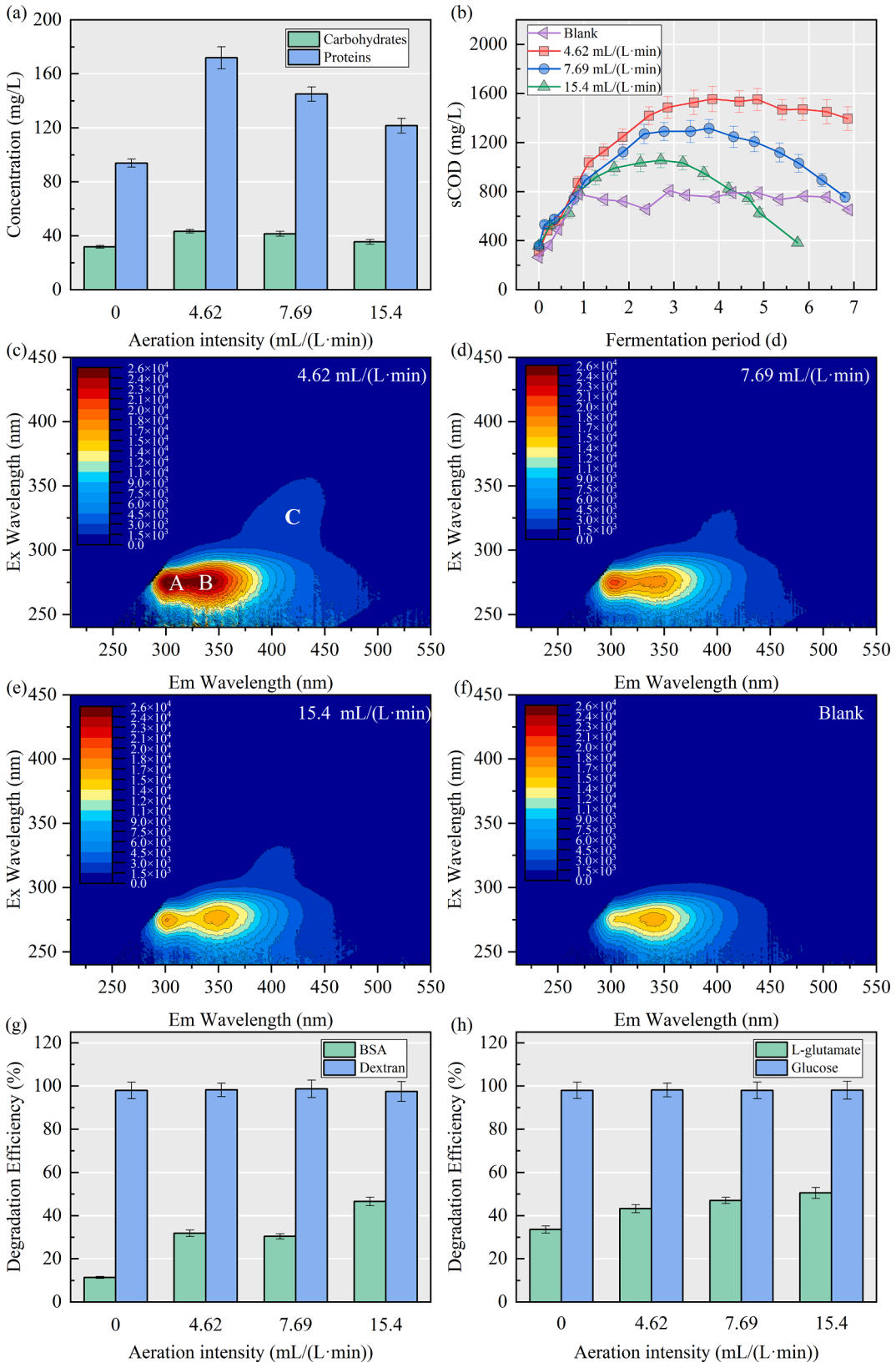


Fig. 3. The effects of micro-aeration intensity with different levels on solubilization, hydrolysis, and acidification: the concentrations of protein and carbohydrate in supernatant after 2 days (a); sCOD during whole periods of anaerobic fermentation (b); the EEM fluorescence spectrum of DOMs at different levels of micro-aeration intensity after 2 days (c)~(f); the degradation of BSA and Dextran after 3-day hydrolysis (g); the degradation of L-glutamate and glucose after 3-day acidification (h). Error bars in (a), (b), (g), and (h) represent standard errors of triplicate samples.

microorganisms are increased [54].

Figure 3(h) illustrates the degradation of L-glutamate and glucose as model hydrolyzed products at the different levels of micro-aeration after three-day fermentation, no significant difference (one-sample *t*-test, $p = 0.277$) was found in the glucose degradation with and without micro-aeration. In contrast, the increase of the micro-aeration intensity contributed to the acidification of L-glutamate. As the micro-aeration intensity increased from 0 to 15.4 mL/(L·min), the degradation efficiency of L-glutamate enhanced from 33.6% to 50.5%. Thus, in the range of 4.62 ~ 15.4 mL/(L·min), the hydrolysis and acidification were both accelerated through micro-aeration assistant. However, the negative effects (facultative consumption of hydrolyzed and acidified products) could be strengthened simultaneously with the increasing aeration intensity, even covering up the positive effect of micro-aeration under over-dose aeration conditions. Hence, precise oxygen supply control is crucial to realize the scale application of micro-aeration in anaerobic fermentation system.

To assess the effects of aerobic respiration on L-glutamate and glucose degradation efficiency, the VFA generated from the above acidification model experiments were also measured. As shown in Fig. S3, the VFA of the unaerated batch increased sharply during days 0 ~ 3 and kept stable during days 3 ~ 4. The VFA in the stable stage of the unaerated batch could be regarded as a reference value owing to the lack of facultative consumption process. The VFA peaks of the micro-aerated batches with 4.62 and 7.69 mL/(L·min) were little different from the reference value, suggesting that the aerobic respiration before the VFA peak could be ignored. In comparison, the VFA peak of 15.4 mL/(L·min) was remarkably lower than the reference value, indicating obvious aerobic respiration at the high aeration intensity conditions. However, on day 2, all the three micro-aerated batches had significantly (one-sample *t*-test, $p = 0.042$) higher VFA than the unaerated batch, which also verified the acidification process can be promoted by micro-aeration. Fig. S3(b) also illustrates that the acetogenic process was beneficially improved with the presence of micro-aeration, which is well consistent with the previous experimental results (Fig. 2(b)–(f)).

3.2. *Influences of aeration cycle time on CEHRAS anaerobic fermentation with intermittent micro-aeration*

3.2.1. *VFAs production and compositions*

According to our pervious batch experiments (Fig. S4), the high acetate proportion fermentate was considered more suitable to serve as carbon source for the heterotrophic denitratation. Taking into account the VFA production and distribution, the influences of aeration cycle time on CEHRAS fermentation with intermittent micro-aeration was further investigated at 7.69 mL/(L·min). As shown in Fig. 4(a), during the rapid accumulation stage (0 ~ 40 h), VFA production performances at the different aeration cycle times (15, 30, 45, and 60 min) were similar. After 40 h, the VFA production of the short aeration cycle (15 min) immediately entered the plateau stage, while the VFA production

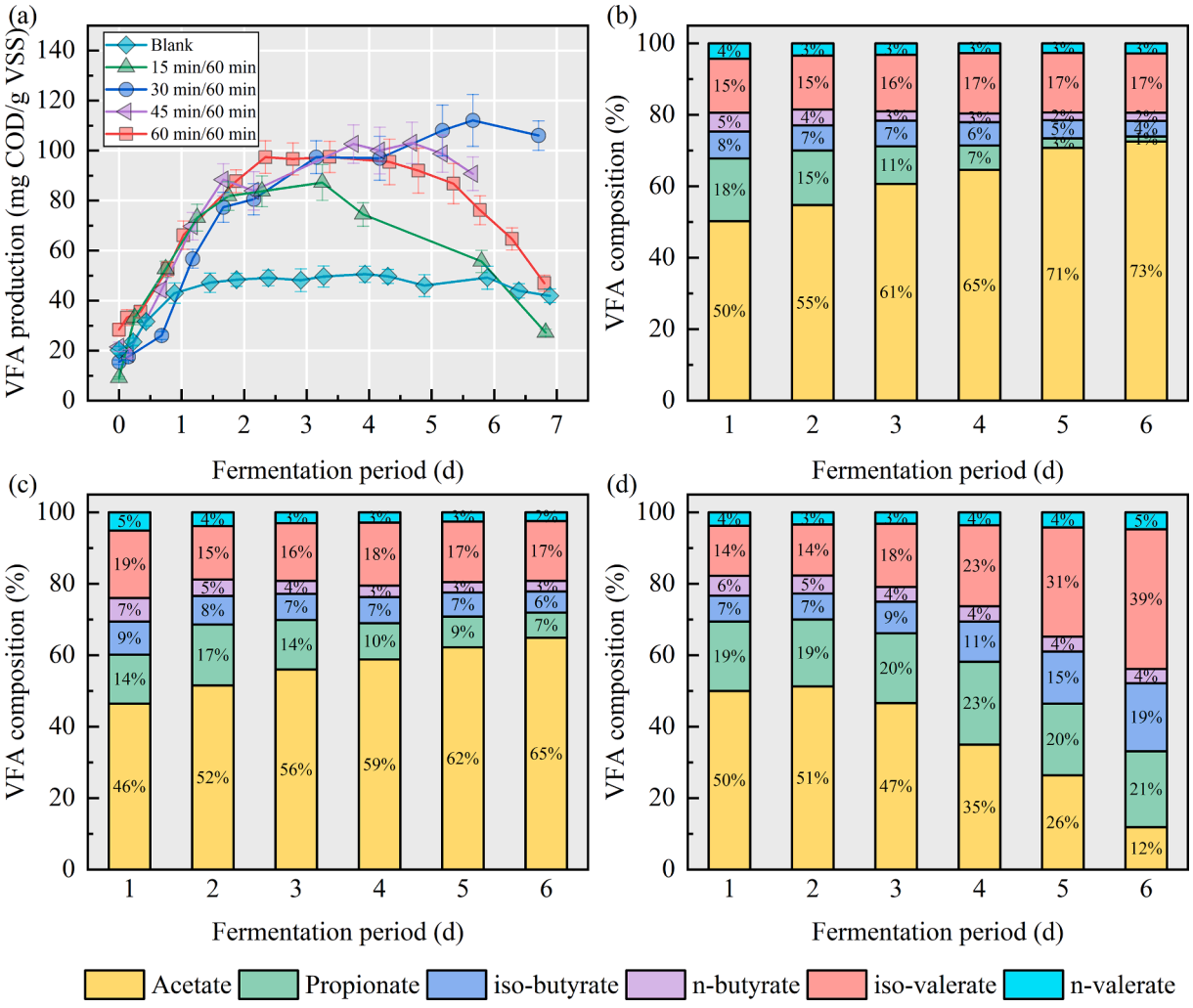


Fig. 4. Total VFAs production profiles with intermittent micro-aeration during whole CEHRAS anaerobic fermentation (a) and VFAs distribution at different aeration cycle time: 45 min/60 min (b), 30 min/60 min (c), and 15 min/60 min (d). Error bars in (a) represent standard errors of triplicate samples.

of the long aeration cycle (45 and 60 min) showed an obvious increasing trend before the plateau stage. In contrast, the VFA production of the medium-aerated cycle (30 min) continuously rose in the whole fermentation period.

The reason for these similar VFA performances in the initial 40 h is that the microorganism tended to degrade carbohydrates rather than the protein in the initial stage of anaerobic fermentation, in which the VFA production was mainly derived from the degradation of carbohydrates. However, it was inferred that micro-aeration had little effect on carbohydrate degradation (section 3.1.2). After 40 h, the main substrate shifted from carbohydrates to proteins with the extended fermentation period. As for the protein-type substrates, the aeration cycle time probably has apparent effects on their hydrolysis process and the aerobic oxidation of their hydrolyzed and acidified products. On the one hand, similar to the aeration intensity, the solubilization and hydrolysis efficiencies of proteins might be positively correlated with the aeration cycle time. Thus, the poor VFA performance of short aeration cycle time (15 min) in the later stage (after 40 h) was attributed to the low efficiency of protein disintegration and hydrolysis. On the other hand, the prolonged aeration cycle time (i.e., 45 min and 60 min) would further

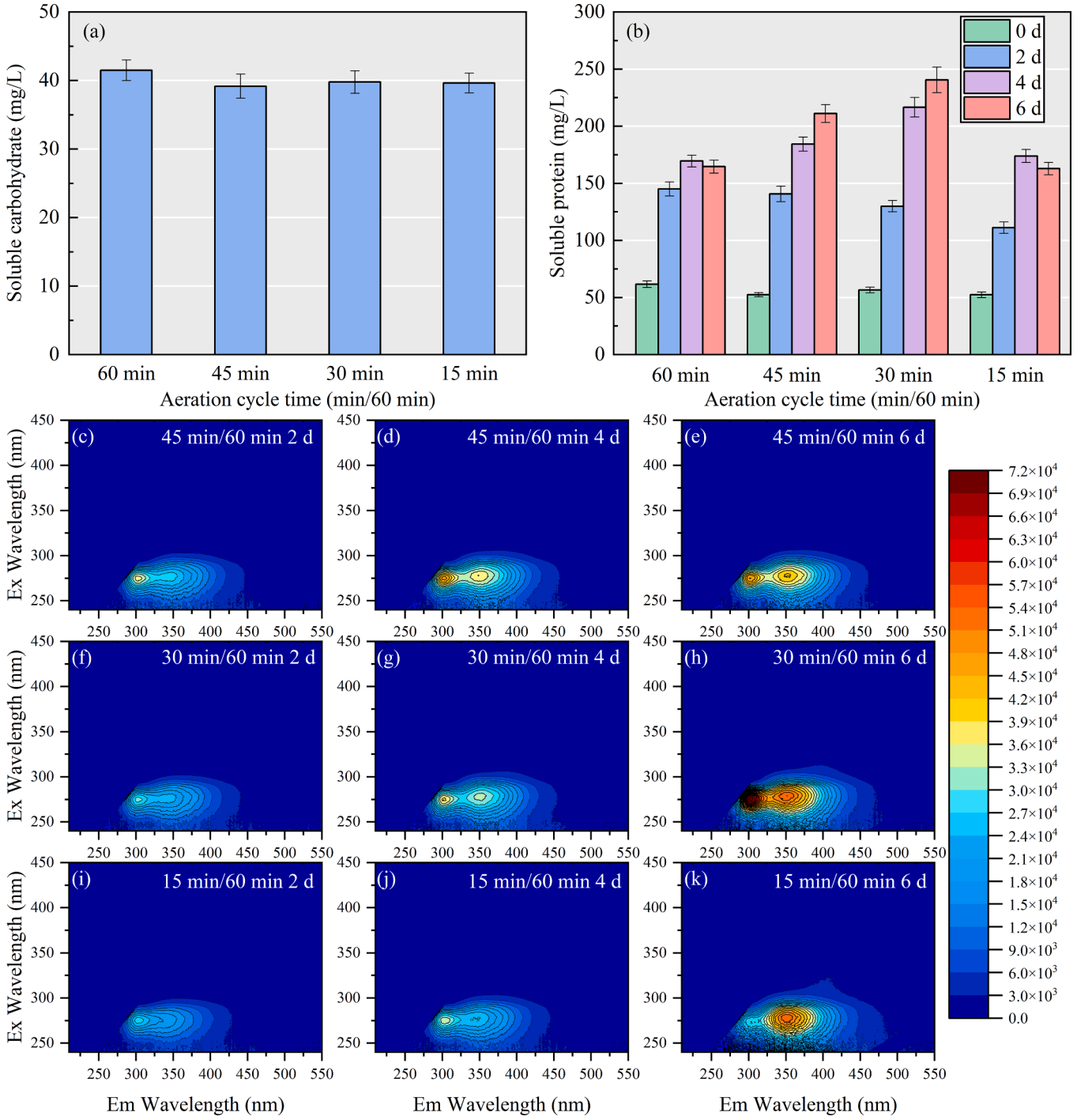


Fig. 5. Carbohydrates in supernatant after 2-day intermittent micro-aeration fermentation (a); variation of protein contents after 0, 2, 4, and 6 days at different aeration cycle time (b); EEM profiles of the soluble organics in the fermenters of CEHRAS with different aeration cycle time after 2, 4 and 6 days (c)~(k). Error bars in (a) and (b) represent standard errors of triplicate samples.

strengthen the aerobic respiration process, resulting in decreased VFA accumulation. The sCOD variation trends at different aeration cycle times in Fig. S5 were consistent with the VFA production. Therefore, the positive and negative effects of aeration cycle time on VFA production coexisted, especially for the protein substrates.

Figure 4(b)~(d) depict the VFA distributions at different aeration cycle times. The abundance of acetate declined with the shortening of the aeration cycle time, while the proportions of other VFAs (i.e., propionate, *iso*-butyrate, and *iso*-valerate) gradually rose in the meantime. The acetate proportions were 70%, 65%, 59%, and 35%, respectively, when the aeration cycle times were 60, 45, 30, and 15 min on day 4. In contrast, *iso*-valerate accounted for 11% (60 min), 17% (45 min), 18% (30 min), and 23% (15 min) of the total VFAs on day 4, respectively. It indicates that the aeration cycle time considerably impacts VFA distribution, especially acetic acid, which is similar to the effects of aeration intensity.

3.2.2. Carbohydrates and proteins

As seen in Fig. 5(a), there was no remarkable difference in the soluble carbohydrates in the first two days of anaerobic fermentation at the aeration cycle time of 60, 45, 30, and 15 min, respectively. The soluble carbohydrates were 41.5, 39.2, 39.8, and 39.6 mg/L, respectively. The soluble proteins in supernatant on 2 d were 145 mg/L (60 min), 141 mg/L (45 min), 130 mg/L (30 min), and 111 mg/L (15 min), respectively (Fig. 5(b)). With fermentation periods prolonged to four days and six days, different variation trend of soluble proteins occurred in above aeration cycle time group (Fig. 5(b)). The soluble proteins of the 30 min/60 min intermittent aeration batch were sharply increased to 217 mg/L and 241 mg/L, respectively, which were much more than that in the continuous, 45 min/60 min, and 15 min/60 min intermittent aeration batches.

As mentioned above, the hydrolysis and acidification efficiencies of carbohydrates were significantly (two-sample *t*-test, Welch correction, all p values < 0.05) higher than those of proteins, which were little affected by the micro-aeration. Hence, the VFA accumulation was mainly originated from the degradation of carbohydrates in the initial fermentation stage, and resulted in the similar soluble carbohydrates in all batches on 2 d. The results of Fig. 5(b) imply the underlying positive relationship (Pearson correlation coefficient = 0.961) between the initial protein release (2 d) and aeration cycle time. Nevertheless, compared to the aeration intensity, the aeration cycle time had less impact on the soluble protein contents (Fig. 3(a) and Fig. 5(b)), indicating the weak influence on the sludge solubilization and hydrolysis during the initial stage. Thus, minor differences were in the VFA accumulation rate of variational aeration cycle times during initial stage.

We attempted introduce the concept of average aeration intensity which could quantify the different aeration cycle time, to clarify the rationales behind the diverse variation trend in soluble proteins of above groups in the later fermentation stage (4 ~ 6 d). Compared to the 15 min/60 min batch, the average aeration intensity was higher in the 30 min/60 min batch. As a result, the higher average aeration intensity promoted the complex polymers to disintegrate and release into the liquid phase as soluble species. The reason for the lower soluble proteins in the 45 min/60 min and continuous aeration batches than those in the 30 min/60 min batch is that the hydrolysis rate of soluble protein could be increased at higher average aeration intensity. However, the amounts of oxidized amino acids could increase simultaneously, which might be the crucial factor causing the reduction of VFA accumulation and sCOD in the later stage (days 4 ~ 6). As discussed above, the solubilization and hydrolysis performance of CEHRAS with the intermittent micro-aeration could be equal to that with the continuous micro-aeration in the initial anaerobic fermentation stage, and the detrimental effects caused by aerobic respiration of facultative anaerobes in the later stage could also be effectively avoided.

3.2.3. EEM fluorescence spectrum

The EEM spectroscopy was applied to determine the characteristic of DOMs at different levels of aeration cycle time. Two protein-like peaks were observed in the EEM spectrum at different levels of aeration cycle time, i.e., peak A (Ex/Em 270/300) and peak B (Ex/Em 275/350), which were ascribed to the tyrosine and tryptophan protein-like substances, respectively [50,51].

In accordance with the results in Fig. 5(b), on day 2, the intensities of peak A and peak B were in the descending order of the 45 min/60 min > the 30 min/60 min > the 15 min/60 min. After adjusting the EEM spectra of continuous/intermittent micro-aeration to the same scale (data not shown), it is found that the division of the intensities of peak A and peak B in intermittent aeration groups was less than that in continuous aeration groups on day 2, which means the aeration intensity has a great impact on disintegration and hydrolysis than aeration cycle time within initial fermentation periods. As prolonged to the day 6, the intensities of peak A and peak B of the 45 min/60 min batch were obviously weaker than those of the 30 min/60 min batch, indicating that the increased aeration cycle time led to the acceleration of protein hydrolysis and amnio acid aerobic oxidation. In addition, only peak B appeared on day 6 in the 15 min/60 min batch, while the peak A of which is not obvious, similar to the EEM spectrum of the unaerated batch in Fig. 3(f). Therefore, the shorter aeration cycle time is considered not conducive to the release of tyrosine protein-like substances.

3.3. Long-term performance of intermittent micro-aeration anaerobic fermentation reactor

The batch fermentation experiments verified that micro-aeration could remarkably enhance the anaerobic fermentation of high-rate activated sludge and increase the acetate proportion in fermentate. Meanwhile, the intermittent micro-aeration was proved to avoid the negative effect caused by the aerobic respiration of facultative anaerobes in the later stage. To further evaluate the long-term operation performance of CEHRAS fermentation with micro-aeration participation, one SBR with intermittently micro-aerated anaerobic fermentation (MaSBR) was implemented for 30 days. In contrast, another SBR without aeration (AnSBR) was also operated for 15 days. Taking into account the VFA production, acetate proportion and economic feasibility, the optimal operation conditions from batch experiments were adopted into the MaSBR and AnSBR, respectively. As summarized on Table 1, the 45 min/60 min of micro-aeration at 7.69 mL/(L·min) had the highest acetate production, though its acetate proportion was lower than the continuous micro-aeration at 7.69 mL/(L·min). Here, we preliminarily believed that the 45 min/60 min of micro-aeration at 7.69 mL/(L·min) was the optimal condition by considering the power consumption of air supply. Thereby, the SRT of MaSBR and AnSBR were 5 days and 2 days, respectively. The total intermittent aeration cycle of MaSBR was 60 min with an aerated cycle of 45 min. Note that a comprehensive techno-economic analysis is needed to evaluate the micro-aeration mode and parameters in the future.

3.3.1. Reactor performance

As shown in Fig. 6(a), the VFA production of AnSBR was stable at 54.3 ± 3.3 mg COD/g VSS after day 2, while the VFA production of MaSBR increased to 104.4 ± 14.2 mg COD/g VSS after day 5, which was accordant with the batch experiment results. The average VFA distribution in the fermentate of MaSBR (days 6 ~ 31) is shown in Fig. 6(b). The average acetate proportion of MaSBR was 59%, which was lower than the batch experiment result (71%). This division was owing to the rise in propionate proportion (from 3% to 15%), which might be attributed to the different feeding modes or the fluctuations in feeding sludge composition, i.e., the increase of the protein contents. In general, acetate and butyrate are the main fermentation products of carbohydrate-rich substrates, while propionate is mainly derived from the degradation of protein species [55,56]. However, compared to

Table 1
Summary of VFA production and acetate proportion on day 4 ~ 5 for each fermentation batch experiments.

Aeration intensity(mL/(L·min))	Aeration cycle time (min)	VFA production (mg COD/g VSS)	Acetate proportion (%)	Acetate production (mg COD/g VSS)	Fermentation time (day)
7.69	60	95.5	70.1%	66.9	4
7.69	60	86.8	77.0%	66.8	5
7.69	45	100	64.6%	64.6	4
7.69	45	101	70.8%	71.5	5
7.69	30	97.0	58.8%	57.0	4
7.69	30	108	62.2%	67.2	5
7.69	15	74.5	35.0%	26.1	4
7.69	15	65.1	26.4%	17.2	5
4.62	60	110	52.9%	58.4	4
4.62	60	104	58.3%	60.7	5

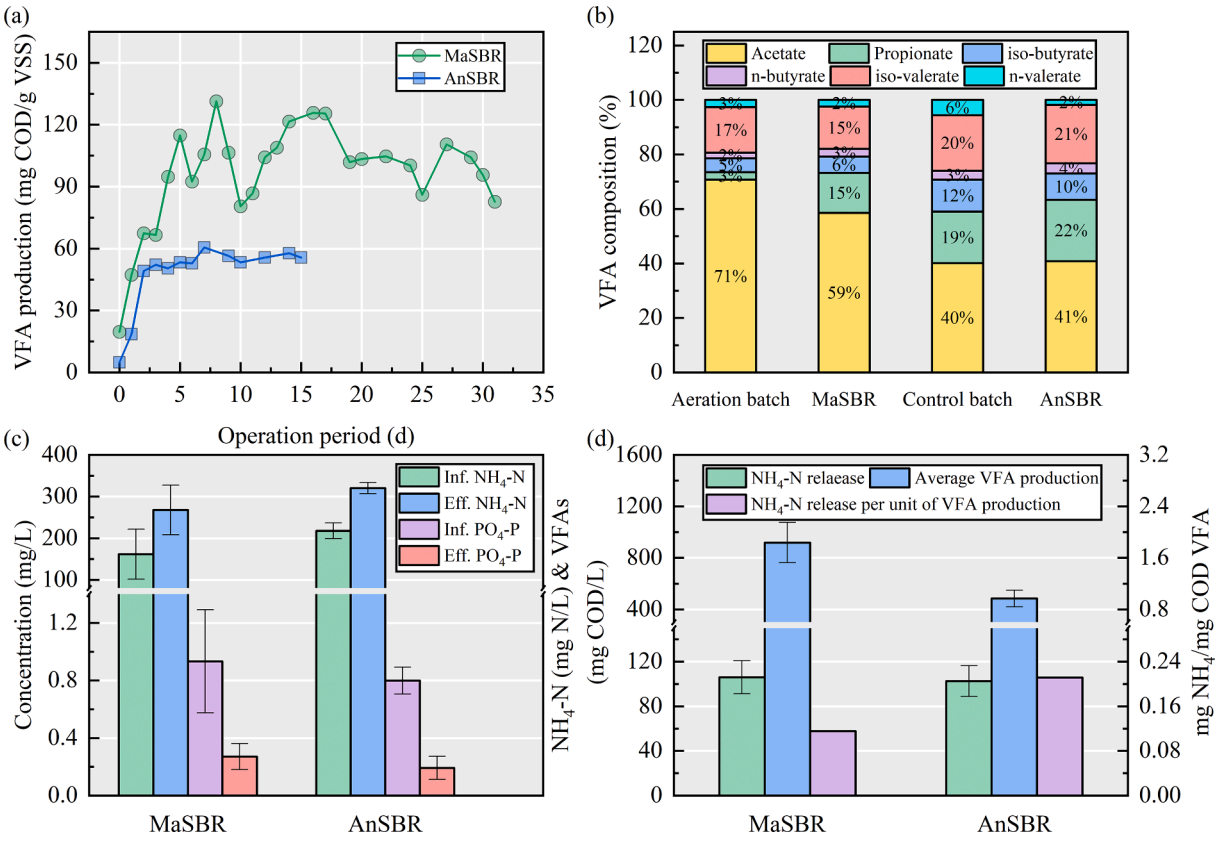


Fig. 6. The performance of long-term VFA production from CEHRAS of MaSBR and AnSBR during 30 days and 15 days (pH 8.0, T = 25 ℃), respectively (a); the average percentage of individual VFA of MaSBR, AnSBR, and the corresponding aeration batch (b); the concentrations of NH_4-N and PO_4-P in the influent and effluent of the MaSBR and AnSBR (c); the NH_4-N release, average VFA production and NH_4-N release per unit of VFA production of MaSBR and AnSBR (d). Error bars in (c) and (d) represent standard errors in quasi-steady states.

AnSBR, there was still a considerable conversion of propionate to acetate in MaSBR, with propionate decreased from 22% to 15% and acetate increased from 41% to 59%. In addition, the ratios of butyrate and valerate in the MaSBR fermentate were consistent with the batch experiment results. It indicates that micro-aeration could still effectively promote the conversion of propionate, butyrate, and valerate to acetate during the long-term operation, although there were composition variations in the feeding sludge. In addition, the VFA distribution of AnSBR fermentate was almost consistent with the batch experiment. The propionate proportion of AnSBR was slightly higher than the batch experiment (22% vs. 19%), which may be also related to the change in protein contents of the feeding sludge.

3.3.2. *Nitrogen and phosphorus releases*

The micro-aerated fermentate of high-rate activated sludge was considered to be a carbon source for the heterotrophic denitratation of the mainstream nitrogen removal process. Considering that the fermentate as a carbon source may increase the mainstream nitrogen and phosphorus loading, NH_4-N and PO_4-P in the influent and effluent of the MaSBR and AnSBR were measured. The influent and effluent NH_4-N of the MaSBR were 162 ± 60 and 268 ± 59 mg N/L, respectively, while those of the AnSBR were 218 ± 19 and 320 ± 14 mg N/L, respectively. The NH_4-N release of the MaSBR and AnSBR were 106 ± 15 and 103 ± 14 mg N/L, respectively, which were not obviously different from the NH_4-N release of the batch experiments in Fig. S6 (121 mg N/L for the micro-aerated batch on day 5, and 94.6 mg N/L for the unaerated batch on day 3). Although the NH_4-N release was similar between the MaSBR and AnSBR, the VFA production of MaSBR was sharply (two-sample *t*-test, Welch correction, $p < 0.001$) increased from 484 ± 64 mg COD/L to 919 ± 156 mg COD/L compared with AnSBR. That is to say, the NH_4-N

release per unit of VFA production was remarkably reduced from 0.212 mg N/mg COD to 0.115 mg N/mg COD (45.8% reduction). The batch experiments already verified that micro-aeration could give the anaerobic fermentation system some nitrification and denitrification capacity (Fig. S6). Besides the traditional nitrification/denitrification, a novel nitrogen removal pathway from anaerobic ammonium oxidation coupled with Fe (III) reduction (i.e., Feammox [57]) might existed in our fermentation systems. Yang et al. [58] demonstrated that continuous nitrogen removal driven by Fe (II)/Fe (III) cycle could be achieved via intermittent aeration (150 mL/min), it is found that the NH_4-N content was only 10.2 mg/L and the TN removal efficiency reached 98.5% on day 90. The direct electron acceptors (e.g., O_2 or Fe^{3+}) and donors (e.g., organics or Fe^{2+}) in nitrification and denitrification of micro-aerated fermentation system are still unclear, more efforts need to be done in the future. Therefore, the nitrogen removal function of MaSBR could explain the division of specific ammonia release between MaSBR and AnSBR. In addition, the CEHRAS contained poor biomass as a result of a short SRT (0.5 days), indicating a lower protein content than the secondary clarifier sludge with a long SRT. It is clear that protein is the primary source of ammonium release. Therefore, the potential for ammonium release during anaerobic fermentation of CEHRAS would be unquestionably lower than that of secondary clarifier sludge. Thus, micro-aerated fermentate of CEHRAS has certain advantages and feasibility as a carbon source for the mainstream heterotrophic denitratation process.

As shown in Fig. 6(c), the PO_4-P in feeding sludge of the MaSBR and AnSBR were very low, 0.933 ± 0.359 mg P/L and 0.799 ± 0.094 mg P/L, respectively, and the PO_4-P in the decanted sludge of these two SBR were 0.270 ± 0.090 mg P/L and 0.192 ± 0.081 mg P/L, respectively. These results indicate no phosphorus release from CEHRAS during anaerobic fermentation, Fe salt in the high-rate activated sludge reactor makes phosphorus in the sludge combine with Fe as iron-phosphorus compounds, which are difficult to dissolve and release [59]. Therefore, CEHRAS anaerobic fermentate as a carbon source for heterotrophic denitratation reactor exhibited additional advantage, i.e., the increase of mainstream phosphorus loading was not required to be considered. More importantly, the Fe-P compounds (such as vivianite) in the sludge could also be considered for recycling and reusing [5,60].

3.3.3. Microbial community composition

The difference of microbial community composition between AnSBR and MaSBR were assessed through 16S rRNA high-throughput sequencing. A total of 1759 operational taxonomic units (OTUs) were obtained based on 501,311 optimized sequences. The alpha diversity found that the Simpson index was little different between MaSBR and AnSBR (two sample *t*-test, $p = 0.111$), which suggested that micro-aeration did not significantly affect the microbial diversity, but may enrich the key functional microorganisms for CEHRAS anaerobic fermentation.

Figure 7 shows the genus-level distribution of microbial communities in the feeding sludge (CEHRAS), AnSBR and MaSBR. The dominant genera in the feeding sludge included *norank_f_Saprospiraceae* (23.8%) and *Acinetobacter* (14.4%), which are common genera in wastewater treatment, and possess a strictly aerobic heterotrophic metabolism [61,62]. Compared with the feeding sludge, the relative abundances of these two genera were remarkably lower in AnSBR and MaSBR, indicating that the both fermentation conditions could decrease aerobic bacteria.

Lots of strict anaerobic hydrolytic and fermentative bacteria such as *Christensenellaceae_R7_group, Paludibacter*, and *Sedimentibacter* [63–65] were enriched in the presence of micro-aeration, which increased from 2.99%, 1.42%, and 0.80% in AnSBR to 4.78%, 3.33%, and 2.07% in MaSBR, respectively. Zakaria et al. [63] reported that *Paludibacter* possessed the ability to degrade carbohydrates for VFAs (mainly acetate and propionate). *Christensenellaceae_R7_group* was proven to produce acetate and butyrate through fermentation of soluble sugars [64]. The genus *Sedimentibacter* was confirmed to be involved in the production of VFA with amino acids, pyruvate, and yeast extract as the main substrates [65]. Interestingly, the abundance of *norank_f_norank_o_OPB41* intensely (two-sample *t*-test, $p < 0.001$) increased from 0.23% (AnSBR) to 4.96% (MaSBR). However, this genus was not detected in the feeding sludge. The *norank_f_norank_o_OPB41* belongs to Class Coriobacteriia.

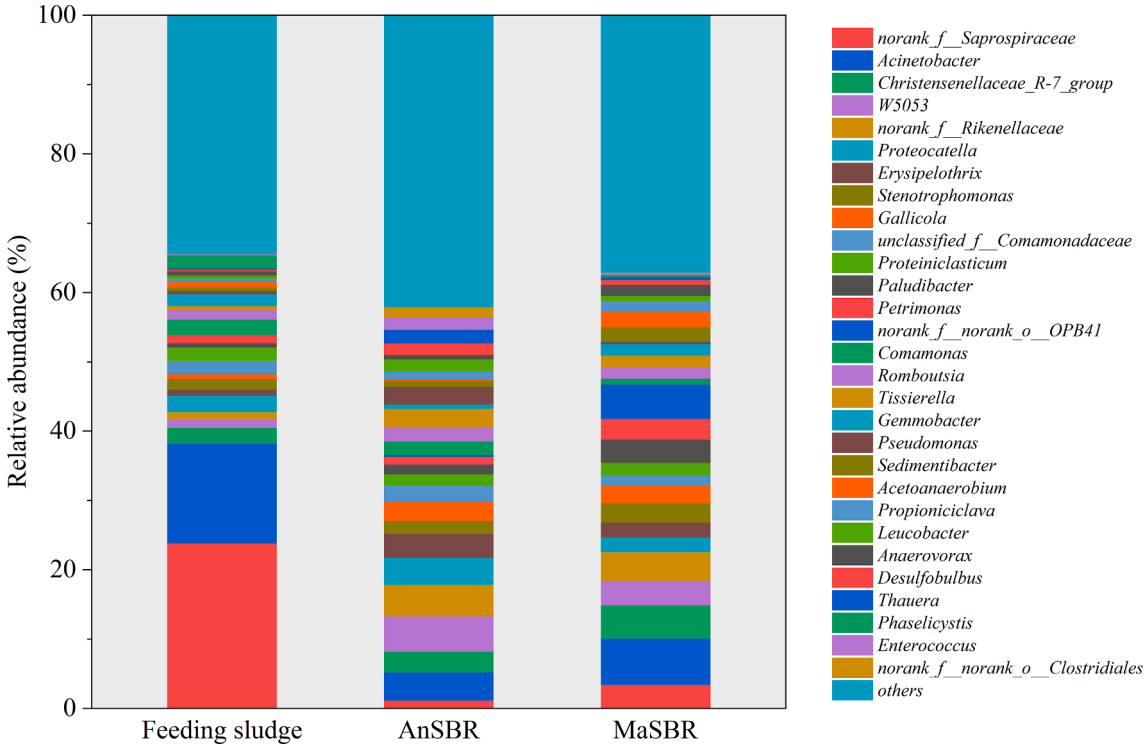


Fig. 7. The distribution of microbial communities in Feeding sludge, AnSBR and MaSBR (on day 15 of fermentation) at genus level.

Accordingly, there was a significant increase in abundance of Coriobacteriia in MaSBR compared with AnSBR at Class level (Fig. S7). The microorganisms under Class Coriobacteriia are strictly or facultatively anaerobic [66]. Baleeiro et al. [67] found that trace O_2 contamination conditions favored the Classes Actinobacteria and Coriobacteriia over Clostridia and Methanobacteia in chain elongation fed with acetate, lactate, and H_2/CO_2. The Class Coriobacteriia contains order Coriobacteriales and order OPB 41, Gupta et al. [66] reported that the order Coriobacteriales produce lactic acid, acetate, formic acid, ethanol, and hydrogen from carbohydrates fermentation. However, the order OPB41 has rarely been discussed in literature. It is worth noting that some fermentative bacteria with acetate as the predominant products were enriched in micro-aeration condition, such as *Acetoanaerobium* (0.28% vs. 2.17%, two-sample *t*-test, $p < 0.001$) and *Anaerovorax* (0.63% vs. 1.62%, two-sample *t*-test, $p < 0.001$), which were identified as homoacetogenesis [68] and acetogenesis microbes [69], respectively.

The reason for the enrichment of strict anaerobic microbes under micro-aeration condition might be that facultative bacteria with high anti-oxidative enzyme activity could remove reactive oxygen species and protect the inner anaerobes [39]. That is to say, micro-aeration can facilitate the constitution of syntrophic relationship among microorganisms. It was found that the abundance of some facultative acidogens (e.g., *Petrimonas*, *Propioniciclava*, and *Gemmobacter*) and aerobic bacteria (e.g., *Stenotrophomonas*) [29,70–72] in MaSBR were higher than those in AnSBR. For instance, *Petrimonas*, a typical VFA and hydrogen producer in fermenters, increased from 1.12% to 3.01% (two-sample *t*-test, $p < 0.001$). Li et al. [29] pointed out the abundance of *Petrimonas* was improved after CaO_2 addition during WAS anaerobic fermentation. CaO_2 could slowly release O_2, H_2O_2, and $Ca(OH)_2$, which may facilitate the formation of reactive oxygen species, thus a similar micro-aeration environment were constructed. Therefore, micro-aeration is conducive to establish microbial communities containing higher activity and abundance of facultative, strict anaerobic and aerobic bacteria, though no obvious distinction in microbial diversity between AnSBR and MaSBR. In addition, according to the previous experimental results, *Erysipelothrix*, *Tissierella*, and *Guggenheimella* were the three dominant genus in CEHRAS fermentation after alkali treatment (pH 10), accounting for 8.97%, 6.88% and 8.38% of the total abundance (Data not shown), indicating that the microbial consortia constituted via micro-aeration tending to more uniform, enrichment and balance due to relative mild-pH condition.

More importantly, no methanogens were detected in all groups in the domain archaea, it is likely that the operational conditions of CEHRAS SBR (low HRT and SRT) suppressed the propagation of archaea species, which means the VFA consumption steps (methanogenesis) could be effectively avoided. Thereby, the CEHRAS exhibited unique advantage and potential as a feedstock substrate in anaerobic fermentation.

3.4. Insights into the application of micro-aerated fermentate for heterotrophic denitratation based on model evaluation

According to the above findings, the micro-aeration treatment has satisfying VFA production and acetate proportion, which was obviously better than that obtained by strictly anaerobic fermentation. Numerous studies have reported that higher nitrite accumulation when acetate serves as the carbon source [16–18,73,74]. Accordingly, it could be inferred that micro-aerated fermentate could be an ideal carbon source for heterotrophic denitrification. We conducted a series of batch denitrification experiments using synthetic and real fermentate to verify this assumption and developed the dual-substrate and two-step denitrification (DSTSN) model (detailed in SI). Based on model fitting and parameter estimation, we predicted the influence of the carbon/nitrate ratio, acetate proportion in fermentate, and the model parameters on heterotrophic denitrification.

3.4.1. The different nitrite accumulation performance between acetate and non-acetate VFA

As illustrated in Fig. 8(a) ~ (j), the synthetic fermentate without acetate exhibited remarkably worse nitrite accumulation than the synthetic fermentate with acetate. At the early stage of the batch experiment, the nitrate reduction rate (0.0848 mg N/L/min) and the nitrite accumulation rate (0.0117 mg N/L/min) of the fermentate without acetate were significantly lower (one-sample *t*-test, $p < 0.001$) than the mean rates of all the synthetic fermentate containing acetate (0.170 and 0.0622 mg N/L/min, respectively) (Table S4). The VFA profiles during denitrification of the real micro-aerated fermentate (Fig. 9) show that the acetate consumption rate was much higher than that of other VFA components. Furthermore, the nitrite accumulation was almost completed in the feast phase of the extracellular carbon source for denitrification. In this feast phase of extracellular carbon sources, the intracellular storage product formation processes on nitrate (proc 1 and proc 2) were dominant in nitrite formation. Meanwhile, intracellular storage product formation processes on nitrite (proc 3 and proc 4) were the main approaches to consuming nitrite. Thus, the difference between the model parameters of the storage processes of S_{Ac} and S_{NAc} could illuminate the distinction in the nitrite accumulation and the extracellular carbon consumption of the batch experiments.

As shown in Table S3, the estimated parameter results for synthetic fermentate show that the saturation constants of acetate and non-acetate ($K_{Ac,P1}$ and $K_{NAc,P2}$) were 20.2 and 48.9 mg COD/L, respectively, indicating a higher affinity for acetate in the nitrite formation processes (proc 1 and 2). Moreover, the storage process rate constant of acetate was 2.67 times higher than non-acetate VFAs ($q_{STO,P1} = 0.738$ mg COD_{STO}/L/min, $q_{STO,P2} = 0.277$ mg COD_{STO}/L/min). The corresponding parameters had a similar tendency in the nitrite consumption processes (proc 3 and 4). Thus, compared with the non-acetate extracellular carbon source, the formation and consumption rates of nitrite and carbon source consumption rate were higher when the intracellular storage product was converted from acetate. Although the storage process rates on nitrate (proc 1 and 2) were lower than that on nitrite (proc 3 and 4), the extracellular carbon source saturation constants ($K_{Ac,P1}$ and $K_{NAc,P2}$) of the former processes were much lower than the later corresponding constants ($K_{Ac,P3}$ and $K_{NAc,P4}$). Consequently, the nitrite formation rate was higher than the consumption rate during the synthesis phase of the intracellular storage product. In this case, both acetate and non-acetate VFAs had the nitrite accumulation phenomenon. Furthermore, the rate ratio of nitrite formation to consumption using acetate was higher than that using non-acetate VFA. Thus, acetate displayed higher nitrite accumulation than non-acetate VFA. In a word, the above experimental data and model parameter values illustrated that acetate was the key VFA in the fermentate of high-rate activated sludge to facilitate nitrite accumulation.

In addition, the peak of internal storage products (X_{STO}) profiles (Fig. 8) increased with the enhancement of the acetate proportion. The peak reflects the amount of the suitable substates for the synthesis stage of intracellular storage product, i.e., the feast phase of the extracellular carbon source, which is the main stage for the nitrite accumulation [20,75]. It means the fermentate with higher acetate proportion triggered satisfactory performance in the heterotrophic denitratation process, which would be discussed in the following.

3.4.2. The decrease of nitrite at the end of denitrification

Figure 8 shows that as the acetate proportion increased in the synthetic fermentate, the maximum nitrite accumulation increased at first (0% to 60%), then remained stable (60% to 80%), and finally slightly declined (80% to 100%). Moreover, under the condition of high acetate proportion, nitrate could be almost entirely depleted during the batch experiment cycle. After the nitrate was used up, the remaining extracellular and intracellular carbon sources as electron donors continued to support the nitrite reduction, which caused the decrease of nitrite accumulation at the later stage of the batch experiments (Fig. 8(d)~(f)).

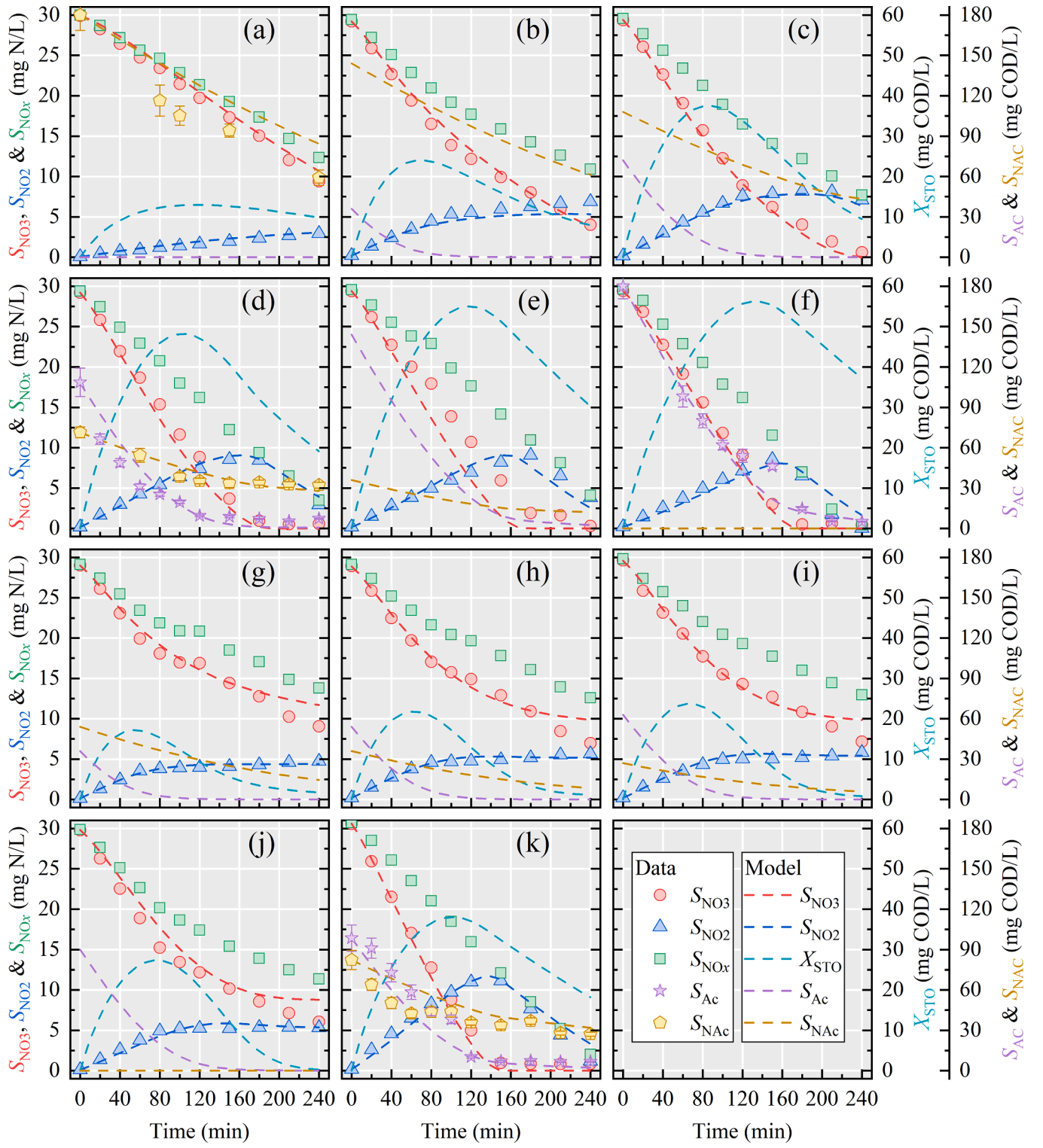


Fig. 8. Analytical data and simulation results of the batch denitrification experiments with different carbon sources, initial carbon/nitrate ratios (C/N) and acetate proportions. synthetic fermentate, C/N = 6 g COD/g N (a)~(f); synthetic fermentate, C/N = 3 g COD/g N (g)~(j); real micro-aerated fermentate, C/N = 6; 0% acetate (a); 20% acetate (b); 40% acetate (c); 60% acetate (d); 80% acetate (e); 100% acetate (f); 40% acetate (g); 60% acetate (h); 70% acetate (i); 100% acetate (j); 60% acetate (k). Error bars of VFAs in (a), (d), (f), and (k) represent standard errors of triplicate samples.

For the batch experiments in which nitrate was not consumed completely, nitrite remained stable after reaching the maximum accumulation. Two conditions caused the incomplete consumption of nitrate: The first was that though the carbon source was enough, the reduction rate of nitrate was slow, and the batch experiment cycle was not enough (Fig. 8(a) and (b)). The second condition was that the carbon source was not enough (Fig. 8(g)~(j)). If only considering the anoxic storage and growth processes, the minimum carbon/nitrate ratio requirement of that nitrate is reduced to nitrite and wholly reduced to dinitrogen are 2.01 and 4.66 mg COD/mg N, respectively (see Eq. S1 and S2 in SI). Therefore, the batch experiments at an initial carbon/nitrate ratio of 3 mg COD/mg N displayed that nitrite could maintain a stable value due to

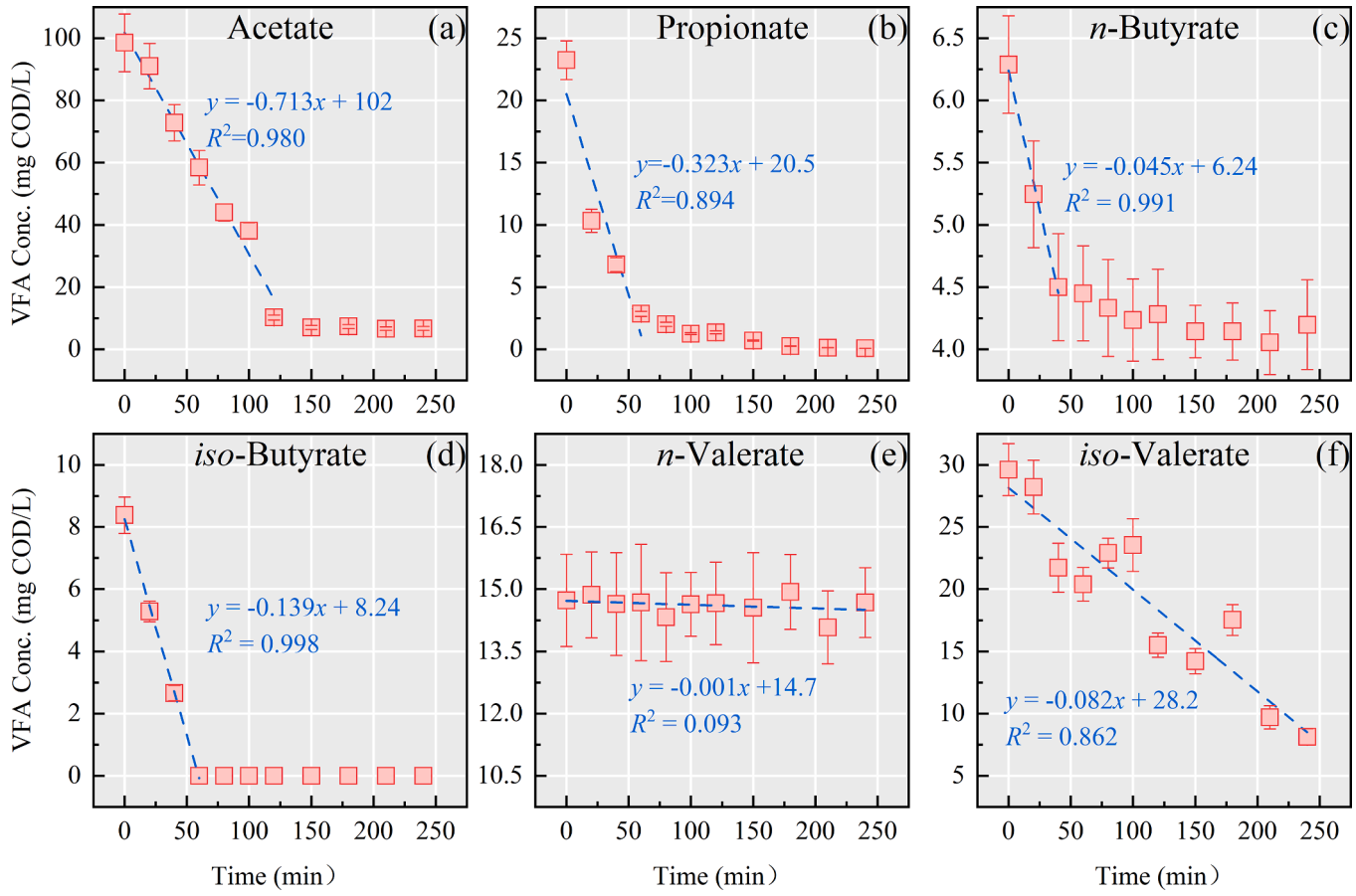


Fig. 9. The VFA component profiles of batch denitrification experiment with real micro-aerated fermentate as carbon source (initial total carbon source/nitrate ratio was 6.0 mg COD/mg N). Error bars represent standard errors of triplicate samples.

the insufficient carbon source (Fig. 8(g)~(j)).

Interestingly, in this study, the activated sludge from the municipal WWTP did not exhibit that nitrate inhibits nitrite reduction, which was reported by Le et al. [17]. Under sufficient acetate conditions (Fig. 8(a)~(f)), the nitrite reduction rate did not change with the decrease of nitrate during the batch experiment cycle. Therefore, the direct factor causing the decrease of nitrite accumulation was that the high carbon/nitrate ratio supported the total nitrate consumption rather than the inhibition by nitrate. However, the heterotrophic denitratation sludge specially cultured by other scholars [17,76] showed the apparent phenomenon that nitrate suppressed nitrite reduction. Also, our specially cultured sludge displayed the similar phenomenon (data not shown). The reason for the different inhibition of nitrite reduction could be that there were significant differences in the genotype or phenotype of the denitrifying functional bacteria of the sludge from municipal WWTP and specially cultured reactor.

3.4.3. The start-up and operation of heterotrophic denitratation reactor

With the parameter values regulated for real fermentate, this study simulated the batch experiments of the sludge from different stages of the start-up and operation of heterotrophic denitratation reactors. The heterotrophic denitratation reactors were fed into different carbon/nitrate ratios and acetate proportions of fermentate. To focus on the mechanisms and simplify the simulations, assumed that during the maturing process of the heterotrophic denitratation sludge, the process rate constants of storage and anoxic growth on nitrate were kept unchanged. In contrast, the corresponding constants on nitrite (i.e., nitrite reduction rate (NiR)) were reduced with the running time.

As illustrated in Fig. 10(a), (d), and (g), when the initial carbon/nitrate ratio was 2.0 ~ 3.0 mg COD/mg N (a range which is mostly adopted in literature), the time that nitrite accumulation increased to 80% of the maximum had a negative correlation with acetate proportion. Moreover, this negative relationship was more pronounced as the cultivation process continued, suggesting that a higher acetate proportion in the fermentate was more conducive to reduce HRT, reduce the reactor size, or increase treatment loading. However, as shown in Fig. 10 (b), (e), and (h), in the initial carbon/nitrite ratio range of 2.0 ~ 3.0 mg COD/mg N, the maximum nitrite accumulation was independent of the acetate proportion, which was in accordance with the experimental results shown in Fig. 8(g), (h), and (i). In contrast, the maximum nitrite accumulation had a positive correlation with the initial carbon/nitrite ratio, which was consistent with Fig. 8(c) vs. (g), (d) vs. (h), and (f) vs. (j). As shown in Fig. 10(c), (f), and (i), the NAR at the time of the maximum nitrite accumulation was positively correlated with the acetate proportion, especially the uncultured and unmatured sludge (100% and 50% NiR rates). For the successfully cultured sludge (10% NiR rates), the nitrite accumulation and the corresponding NAR could maintain > 25 mg N/L and > 90%, respectively, when the initial carbon/nitrate ratio was 2.5 ~ 3.0 mg COD/mg N. And the fluctuation of the acetate proportion did not result in a wide variation in both the nitrite accumulation and the corresponding NAR. Fig. S8 illustrates that regardless of which stage of the start-up and operation of the heterotrophic denitratation reactor, the duration of nitrite remaining ≥80% of the maximum was clearly positively correlated with the acetate proportion, indicating that a higher acetate proportion is more beneficial to the time control of heterotrophic denitratation.

Above experimental data, model fitting, and prediction results revealed the significant roles of increasing acetate proportion in fermentate: (1) shorten reaction time; (2) increase NAR; and (3) extend nitrite accumulation duration. Thus, in light of the function of

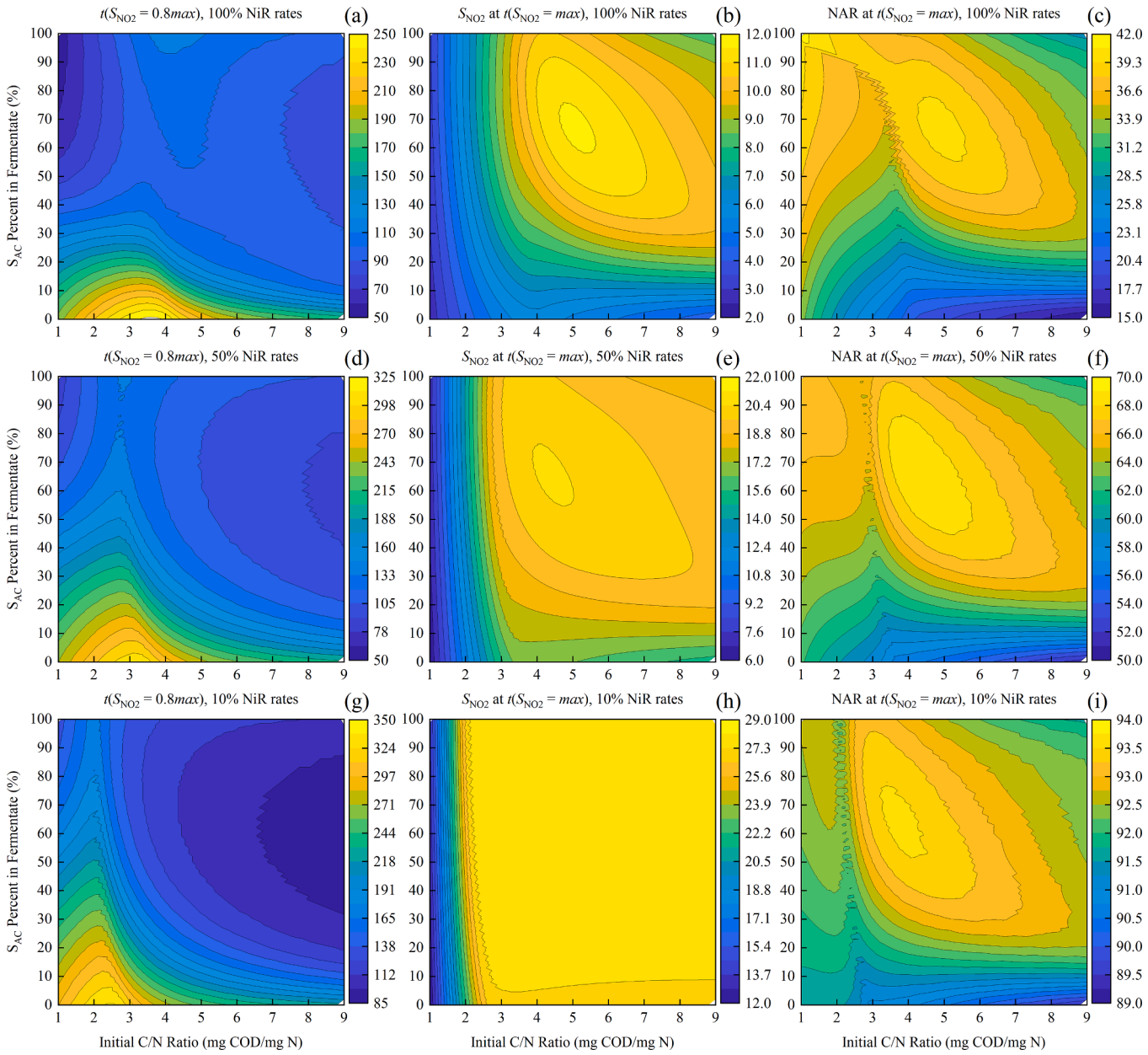


Fig. 10. Denitratation performance variation of the heterotrophic denitratation reactor with different initial carbon source/nitrate ratio and acetate proportions. The reaction cycle time reaching 80% of the maximum nitrite accumulation ((a), (d) and (g)); the maximum nitrite accumulation ((b), (e), and (h)); the nitrite accumulation ratio at the reaction cycle time when nitrite accumulation reaching the maximum ((c), (f), and (i)). Using the relative nitrite reduction activity with respect to the initial value to present the maturity of denitratation sludge.

enhancing acetate production and proportion, the micro-aeration is of great significance for the application of anaerobic fermentation with high-rate activated sludge as substrates in heterotrophic denitratation.

4. Conclusion

Micro-aeration is an effective technology to enhance VFA production and acetate proportion of anaerobic fermentation with CEHRAS as substrates. The aeration intensity and cycle time are two crucial factors for regulating multiple bioprocesses (i.e., hydrolysis, acidogenesis, acetogenesis, and aerobic oxidation) in the micro-aerated fermentation of CEHRAS. High intensities of continuous micro-aeration inevitably cause high-level aerobic respiration of hydrolyzed and acidified products. The intermittent aeration can effectively limit aerobic oxidation in the later fermentation stage. Micro-aeration can enhance the hydrolysis, acidogenesis, and acetogenesis steps of proteins, whereas no obvious effect on the degradation of carbohydrates. The actual long-term operation of CEHRAS anaerobic fermentation with micro-aeration assistant is feasible and economical. The addition of micro-aerated fermentate into the mainstream line of WRRFs has lower increments of nitrogen and phosphorus loadings than that of the regular anaerobic fermentate. The batch denitrification tests and DSTSN model evaluation disclosed that acetate is the crucial component in VFAs to trigger high nitrite accumulation. Higher acetate proportion can shorten the reaction time, increase nitrite accumulation ratio, and extend nitrite accumulation duration. Therefore, micro-aerated fermentate is conducive to successful denitratation reactor operation, especially in the start-up stage.

CRediT authorship contribution statement

Yexuan Wen: Conceptualization, Investigation, Methodology, Data curation, Visualization, Formal analysis, Writing – original draft. **Haixin Jiang:** Conceptualization, Methodology, Software, Visualization, Writing – review & editing. **Ruibo Qian:** Methodology. **Shiting Liu:** Funding acquisition. **Xianchun Tang:** Project administration. **Weiping Huang:** Resources. **Hongbin Chen:** Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2022.136894.

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